

SQL y Álgebra Relacional

1. Sección 1: SQL $\rightarrow$ Álgebra Relacional

1. Vuelos cuyo destino es New Delhi.

SQL:

```
1 SELECT *
2 FROM flight
3 WHERE dest = 'New Delhi';
```

Álgebra Relacional:

$$\sigma_{dest='New\ Delhi'}(flight)$$

2. Nombres de pasajeros con al menos una reserva.

SQL:

```
1 SELECT DISTINCT p.pname
2 FROM passenger p
3 JOIN booking b ON b.pid = p.pid;
```

Álgebra Relacional:

$$\pi_{pname}(passenger \bowtie booking)$$

3. Vuelos que operan a las 16:00 tanto el 2020-12-01 como el 2020-12-02.

SQL:

```
1 SELECT *
2 FROM flight
3 WHERE fdate = DATE '2020-12-01' AND time = '16:00'
4 INTERSECT
5 SELECT *
6 FROM flight
7 WHERE fdate = DATE '2020-12-02' AND time = '16:00';
```

Álgebra Relacional:

$$\sigma_{fdate='2020-12-01'\wedge time='16:00'}(flight) \cap \sigma_{fdate='2020-12-02'\wedge time='16:00'}(flight)$$

4. Pasajeros sin reservas (pid y nombre).

SQL:

```
1 SELECT p.pid, p.pname
2 FROM passenger p
3 WHERE p.pid NOT IN (SELECT pid FROM booking);
```

Álgebra Relacional:

$$\pi_{pid,pname}(passenger) - \pi_{pid,pname}(passenger \bowtie booking)$$

5. Pasajeros masculinos asociados a la agencia 'Jet'.

SQL:

```
1 SELECT DISTINCT p.pid, p.pname, p.pcity
2 FROM passenger p
3 JOIN booking b ON b.pid = p.pid
4 JOIN agency a ON a.aid = b.aid
5 WHERE p.pgender = 'Male' AND a.aname = 'Jet';
```

Álgebra Relacional:

$$\pi_{pid,pname,pcity}(\sigma_{pgender='Male'\wedge aname='Jet'}(passenger \bowtie booking \bowtie agency))$$

2. Sección 2: Álgebra Relacional $\rightarrow$ SQL

1. $\sigma_{src='Chennai' \wedge dest='New Delhi'}(flight)$

SQL Equivalente:

```
1 SELECT *
2 FROM flight
3 WHERE src = 'Chennai' AND dest = 'New Delhi';
```

2. $\pi_{fid}(\sigma_{pid=123}(booking) \bowtie \sigma_{dest='Chennai'}(flight))$

SQL Equivalente:

```
1 SELECT b.fid
2 FROM booking b
3 JOIN flight f ON b.fid = f.fid
4 WHERE b.pid = 123 AND f.dest = 'Chennai';
```

3. $\pi_{aname}(agency \bowtie_{agency.acity=passenger.pcity} \sigma_{pid=123}(passenger))$

SQL Equivalente:

```
1 SELECT a.aname
2 FROM agency a
3 JOIN passenger p ON a.acity = p.pcity
4 WHERE p.pid = 123;
```

4. $(\sigma_{fdate='2020-12-01' \wedge time='16:00'}(flight)) \cup (\sigma_{fdate='2020-12-02' \wedge time='16:00'}(flight))$

SQL Equivalente:

```
1 SELECT *
2 FROM flight
3 WHERE fdate = '2020-12-01' AND time = '16:00'
4 UNION
5 SELECT *
6 FROM flight
7 WHERE fdate = '2020-12-02' AND time = '16:00';
```

5. $\pi_{aname}(agency) - \pi_{aname}(agency \bowtie \sigma_{pid=123}(booking))$

SQL Equivalente:

```
1 SELECT aname
2 FROM agency
3 EXCEPT
4 SELECT a.aname
5 FROM agency a
6 JOIN booking b ON a.aid = b.aid
7 WHERE b.pid = 123;
```